UK Maths Trust

Intermediate Mathematical Challenge

Solutions 2026

© 2026 UK Mathematics Trust

proudly sponsored by [XTX] MARKETS

For reasons of space, these solutions are necessarily brief.

There are more in-depth, extended solutions available on the UKMT website, which include some exercises for further investigation:

www.ukmt.org.uk

supported by Overleaf

1. E The values of the expressions are, respectively, $\frac{3}{2} = 1\frac{1}{2}$; $\frac{6}{3} = 2$; $\frac{10}{4} = \frac{5}{2} = 2\frac{1}{2}$; $\frac{15}{5} = 3$; $\frac{21}{6} = \frac{7}{2} = 3\frac{1}{2}$. Therefore $\dfrac{1+2+3+4+5+6}{6}$ is the greatest.

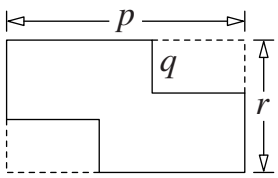
2. D As $2026 \div 60 = 33$ remainder 46, 2026 minutes is equal to 33 hours 46 minutes, that is 1 day, 9 hours and 46 minutes. Therefore at 2026 minutes after 20:26 on Monday it will be 9 hours 46 minutes after 20:26 on Tuesday, that is 46 minutes after 05:26 on Wednesday.
So the required time is 06:12 on Wednesday.

3. D Let the lengths of the sides of the cuboid, in cm, be $n-1$, n and $n+1$.
Then $(n-1) \times n \times (n+1) = 120$, that is $n^3 - n - 120 = 0$. By inspection, we see that $n^3 - n - 120$ is negative when $n \leq 4$, is equal to 0 when $n = 5$ and is positive when $n \geq 6$. Therefore, as the side lengths of the cuboid are positive integers, the cuboid measures, in cm, 4 by 5 by 6.
Hence the area, in cm^2, of each of the largest faces of the cuboid is 5×6, that is 30.

4. E Note that the ratio $1 : \frac{1}{2} : \frac{1}{4}$ is equivalent to the ratio $4 : 2 : 1$.

Now as $4 + 2 + 1 = 7$ and $35 \div 7 = 5$, the smallest part is $1 \times \frac{35}{7}$, that is 5.

5. A In the diagram, in each of the small rectangles, the length of the two solid lines equals that of the dotted lines. Hence the perimeter of the octagon is equal to the perimeter of a rectangle of length p and width r, that is $2p + 2r$.



6. D We need to solve the inequality $\frac{n}{n+1} > 0.92$, that is $n > 0.92n + 0.92$.

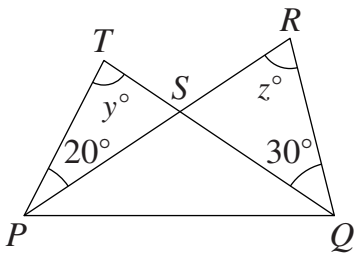
So $100n > 92n + 92$, that is $8n > 92$. Hence $n > 11.5$.
Therefore the first term of the sequence to exceed 0.92 is $\frac{12}{13}$.

7. B Consider the five larger numbers. First note that 97 is the only prime amongst them, so must be paired with 1. Secondly, as 95 is the only multiple of 5, it must be paired with 5. Next, we note that 96 and 99 are both multiples of 3, but 99 is the only multiple of 9 so 96 is paired with 3 and 99 with 9. Finally, this leaves 98, which is 7×14, to be paired with 7.

The five values obtained by dividing each larger number by its partner are $\frac{95}{5} = 19$;
$\frac{96}{3} = 32$; $\frac{97}{1} = 97$; $\frac{98}{7} = 14$; $\frac{99}{9} = 11$.

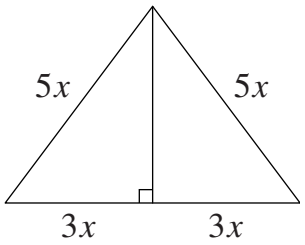
Therefore the required difference is $97 - 11 = 86$.

8. A In a triangle, an exterior angle is equal to the sum of the two interior and opposite angles. Therefore, considering triangle PST, $\angle PSQ = y° + 20°$. Similarly, considering triangle QRS, $\angle PSQ = z° + 30°$.
Hence $y + 20 = z + 30$. So $y = z + 10$.



9. D $5^2 - 1 = 25 - 1 = 24$; $7^2 = 49 - 1 = 48 = 2 \times 24$; $9^2 - 1 = 81 - 1 = 80 = 3\frac{1}{3} \times 24$;
$11^2 - 1 = 121 - 1 = 120 = 5 \times 24$; $13^2 - 1 = 169 - 1 = 168 = 7 \times 24$.
So exactly four of the expressions are divisible by 24. The only exception is $9^2 - 1$.

10. B Let the lengths of the sides of the triangle, in cm, be $5x$, $5x$ and $6x$. The diagram shows that this triangle may be divided into two congruent right-angled triangles which have base $3x$ and hypotenuse $5x$. Therefore, these are both '3, 4, 5' triangles and the perpendicular height of the original triangle is $4x$.



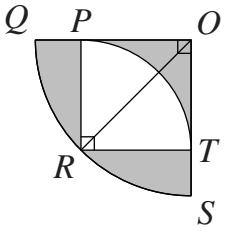
The area of the given triangle is 108 cm^2. Hence $\frac{1}{2} \times 6x \times 4x = 108$. So $12x^2 = 108$, that is $x^2 = 9$. Therefore, as x is positive, $x = 3$ and the length, in cm, of the longest side of the triangle is $6x$, that is 18.

11. B First note that the units, tens and hundreds columns all involve the addition of two single digits and therefore any carried digit from one column to another is at most 1. As K and I represent different digits, K + I ≠ 0. Therefore, in the hundreds column, K + I + any carry is equal to 10. Hence there is a carry of 1 from the hundreds column to the thousands column and we can deduce that U = 2 − 1 = 1.

U	K	M	T
+	I	M	C
2	0	2	6

There cannot be a carry of 1 from the units column to the tens column as that would result in an odd number of tens when added to M + M. So we can deduce that T + C = 6 and M + M = 2 or M + M = 12, that is M = 1 or M = 6. However, the letters all represent different digits and U = 1, so M = 6. Hence there is a carry of 1 from the tens column to the hundreds column and we can therefore deduce that K + I = 10 - 1 = 9.
So T + I + C + K = (T + C) + (K + I) = 6 + 9 = 15.

12. **E** Consider triangle ORT. By Pythagoras' Theorem,
$OR^2 = RT^2 + TO^2 = (2x)^2 + (2x)^2 = 4x^2 + 4x^2 = 8x^2$.
Therefore, as OR is the radius of quarter circle OQS, the area of this quarter circle is $\frac{1}{4} \times \pi \times (OR)^2 = \frac{1}{4} \times \pi \times 8x^2 = 2\pi x^2$.

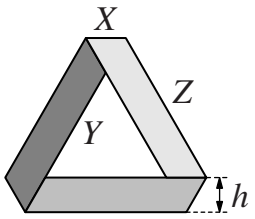


Now the radius of quarter circle PRT is $2x$. Therefore its area is $\frac{1}{4} \times \pi \times (2x)^2 = \frac{1}{4} \times \pi \times 4x^2 = \pi x^2$.
So the total area of the three shaded regions is $2\pi x^2 - \pi x^2 = \pi x^2$.

13. **E** If x is the mean of the five numbers, then $\frac{1 + 2 + 4 + 13 + x}{5} = x$. So $20 + x = 5x$.
Hence $4x = 20$, that is $x = 5$. If x is the median of the five numbers then the numbers in ascending order are 1, 2, x, 4, 13, so, since x is an integer, it is 2, 3 or 4. Finally, if x is the mode of the numbers then it is 1, 2, 4 or 13. Therefore x has six possible values, namely 1, 2, 3, 4, 5 or 13.

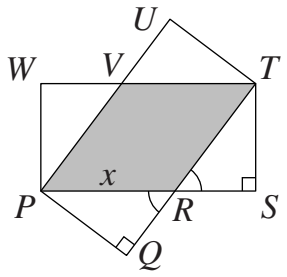
14. **C** As triangle PQR is acute-angled, we need consider only the triangular numbers which are less than 90. These are 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66 and 78. The largest angle in the triangle is greater than 60°, so is either 66° or 78°. If the largest angle is 66°, then the sum of the other two angles is $(180 - 66)°$, that is 112°. The larger of the two remaining angles is then at least $(112 \div 2)°$, that is 56°. So it is also 66° and the remaining angle is $(180 - 2 \times 66)°$, that is 48°. However, 48 is not a triangular number, so it can be deduced that the largest angle in the triangle is 78° and one of the other two angles is at least $((180 - 78) \div 2)°$, that is 51°. So a second angle in the triangle is either 55°, 66° or 78°. This gives three possibilities for the angles of the triangle: 78°, 55°, 47°; 78°, 66°, 36° or 78°, 78°, 24°. As 36 is a triangular number, but 47 and 24 are not, it can be deduced that the smallest angle in the triangle is 36°.

15. **D** Using the formula 'area = $\frac{1}{2}ab \sin C$', the area of the enclosed triangle is $\frac{1}{2} \times Y \times Y \sin 60°$, that is $\frac{1}{2} \times (4X)^2 \sin 60° = 8X^2 \sin 60°$.
Let the perpendicular height of each of the three identical parallelograms be h. Then $\sin 60° = \frac{h}{X}$, so $h = X \sin 60°$. The area of each of these parallelograms is $Z \times h$, so the total area of the three parallelograms is $3Zh$, that is $3 \times 5X \times X \sin 60° = 15X^2 \sin 60°$.
So the required ratio is $8X^2 \sin 60° : 15X^2 \sin 60°$, that is 8 : 15.



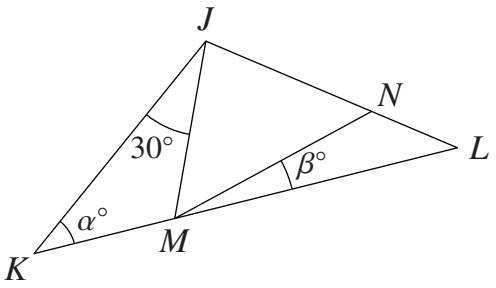
16. **C** Note that 2:15 pm is 75 minutes later than 1 pm. Therefore the distance between Kerry and Karen reduced by 100 km in 75 minutes. So in the 60 minutes between midday and 1 pm, the distance between them reduced by $\left(100 \times \frac{60}{75}\right)$ km, that is $\left(100 \times \frac{4}{5}\right)$ km = 80 km.
Therefore, as they were 200 km apart at 1 pm, Kerry and Karen live 280 km apart.

17. **A** Let P, Q, R, S, T, U, V and W be the points shown. Also, let the length of PR be x. Consider triangles PQR and TSR:
$\angle PQR = \angle TSR = 90°$; $\angle QRP = \angle SRT$ (vertically opposite angles) and $PQ = TS = 1$. Hence the two triangles are congruent (AAS) and $QR = SR = 2 - x$. In triangle PQR, by Pythagoras' Theorem, $PR^2 = PQ^2 + QR^2$. Therefore, $x^2 = 1^2 + (2 - x)^2$, that is $x^2 = 1 + 4 - 4x + x^2$. So $4x = 5$, that is $x = \frac{5}{4}$.



Hence the shaded region is a parallelogram of base $\frac{5}{4}$ and height 1.
So its area is $\frac{5}{4} \times 1 = \frac{5}{4}$.

18. **C** Let $\angle JKM = \alpha°$ and $\angle NML = \beta°$. In triangle JKL, $JK = JL$ so it is isosceles and $\angle JLK = \angle JKL = \alpha°$. Consider triangle NML: as the exterior angle of a triangle is equal to the sum of the two interior and opposite angles, $\angle MNJ = \angle MLN + \angle NML = (\alpha+\beta)°$.

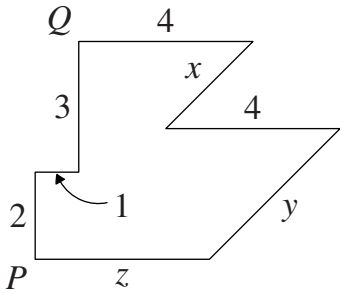


Hence, as JMN is an isosceles triangle in which $JN = JM$, $\angle JMN = \angle MNJ = (\alpha + \beta)°$. So $\angle JML = (\alpha + \beta)° + \beta° = (\alpha + 2\beta)°$.
Consider triangle JKM: once again, using the exterior angle theorem, $(\alpha + 2\beta)° = \alpha° + 30°$.
So $\angle NML = \beta° = 30° \div 2 = 15°$.

19. **D** $1 + \dfrac{16}{1 + \dfrac{1}{1 + \dfrac{1}{x}}} = 11 \quad \Leftrightarrow \quad \dfrac{16}{1 + \dfrac{1}{1 + \dfrac{1}{x}}} = 10 \quad \Leftrightarrow \quad 1 + \dfrac{1}{1 + \dfrac{1}{x}} = \dfrac{16}{10} = \dfrac{8}{5} \quad \Leftrightarrow \quad \dfrac{1}{1 + \dfrac{1}{x}} = \dfrac{3}{5}.$

Therefore $1 + \frac{1}{x} = \frac{5}{3} \quad \Leftrightarrow \quad \frac{1}{x} = \frac{2}{3} \quad \Leftrightarrow \quad x = \frac{3}{2}$.

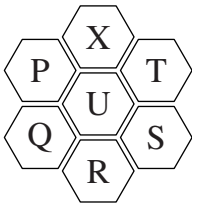
20. **B** Let x, y and z be the lengths shown in the diagram.
Considering the vertical displacements between P and Q, $y \sin 45° + x \sin 45° = 2 + 3$. So $(x + y) \sin 45° = 5$.
Hence $x + y = 5 \div \sin 45° = 5 \div \frac{1}{\sqrt{2}} = 5\sqrt{2}$. Considering the horizontal displacements going anticlockwise from P around the figure, $z + y \cos 45° - 4 + x \cos 45° - 4 - 1 = 0$.
Therefore $z + (x + y) \cos 45° = 9$.



So, as $(x + y) \cos 45° = (x + y) \sin 45° = 5$, $z + 5 = 9$, that is $z = 4$.
Hence the perimeter of the figure is $z + y + 4 + x + 4 + 3 + 1 + 2 = z + (x + y) + 14 = 18 + 5\sqrt{2}$.

21. **A** We are given that $x + y = 6$ and $x^3 + y^3 = 198$.
Note that $(x + y)^3 = x^3 + 3x^2y + 3yx^2 + y^3$.
Hence $3xy(x + y) = (x + y)^3 - (x^3 + y^3)$.
So $3xy \times 6 = 6^3 - 198 = 216 - 198 = 18$.
Therefore $xy = 18 \div 18 = 1$.

22. **C** At some point, the bee must go to the inner cell U. Both before that and then again after that, it must either proceed clockwise or anticlockwise round the outer cells. Consider in turn the routes that reach U directly after each of the outer cells. If immediately after X, then there are only two possible routes, XUPQRST or XUTSRQP. If immediately after P, the bee will have started via the routes XPU or XTSRQPU. The route XTSRQPU is already complete, and XPU can continue QRST or TSRQ. This gives three routes; and the same will apply for routes approaching U directly after one of Q, S and T.



Finally, there are four routes approaching U from R, namely XPQRUTS, XPQRUST, XTSRUPQ or XTSRUQP So, in total, the number of possible circuits is 2 + 4 x 3 + 4, which is 18.

23. **B** Note that Oliver only had 1% saying no. So he asked some multiple of 100 people. Let this number be $100n$, of whom n said no. Let x be the number of people Teresa asked. Since, overall, 2% said no, then the overall number who said no was $(100n + x) \times 2/100$, that is $2n + (x/50)$. So x must be a multiple of 50 and the number saying no to Teresa is then $n + x/50$. This is minimal when $n = 1$ and $x = 50$, with 2 people saying no to Teresa.

24. **B** As shown in figure (i), the line segment from the vertex of a triangle to the midpoint of the opposite side divides the triangle into two triangles which are equal in area. This is because the two smaller triangles have bases which are equal in length and they share the same perpendicular height, h.

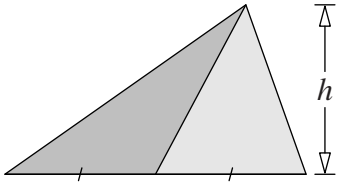


Figure (i)

As K, L, M and N are the midpoints of their respective sides, figure (ii) shows how the above property may be used to divide the area of quadrilateral $PQRS$ into four pairs of triangles with each pair having the same area. Let these areas be A, B, C and D, as shown.
The area of quadrilateral $TMSN$ is 15, so $A + D = 15$.
Similarly, $B + C = 20$ and $C + D = 24$.
Therefore, $A + D + B + C - (C + D) = 15 + 20 - 24$, that is $A + B = 11$.

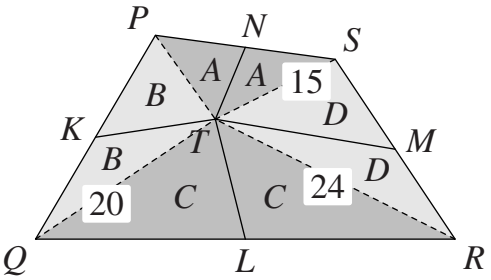


Figure (ii)

So the area of quadrilateral $NPKT$ is 11.

25. **C** When Zoe chooses three days of the week to go swimming, there are three gaps between these days, i.e. between the first and second swimming days, between the second and third and between the third and first. However, a gap may be zero days. For example, if Zoe chooses Monday, Tuesday and Thursday, the gaps, in days, are 0, 1 and 3 respectively.
The sum of the numbers representing the gaps is 7 − 3, that is 4. Therefore, another possibility for the three gaps is (0, 0, 4) which would correspond, for example, to the days chosen being Tuesday, Wednesday and Thursday. This combination of gaps could occur in two other ways, namely (0, 4, 0) and (4, 0, 0). As has been shown, a different possibility for the gaps is (0, 1, 3) and this combination could occur in five other different ways, namely (0, 3, 1), (1, 0, 3), (1, 3, 0), (3, 0, 1) and (3, 1, 0).
The only other possible combinations of gaps are (0, 2, 2) and (1, 1, 2). As with (0, 0, 4), both of these can be chosen in three different ways. Therefore, the gaps between the days on which Zoe chooses to swim may occur in a total of fifteen equally likely ways and in exactly nine of these her schedule will include at least three days in a row without a swim.

So the required probability is $\frac{9}{15}$, that is $\frac{3}{5}$.